Viju Sanjai

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Education

Vellore Institute of Technology Chennai, TN
Bachelor of Technology in Electronics and Communication (GPA: 8.16) *Aug 2023 – Present*

Technical Skills

Languages: Python, JavaScript/TypeScript, Java, C/C++, Rust, C#, SQL, HTML/CSS
Frameworks: React, Next.js, Express, Flask, .NET, Tauri, WebGPU, Hono
Developer Tools: Git, GitHub Actions, Docker, AWS, Cloudflare, Supabase, VS Code, Linux/GDB, Figma
Concepts/Areas: Full-Stack Development, Systems Architecture, Kernel Development, Reverse Engineering, Debugging, Unit Testing, CI/CD, Problem Solving

Experience

Lagoon Technologies Chennai, TN
Software Development Intern *Sep 2024 – Feb 2025*

- Engineered and optimized internal enterprise software tools in **C# and .NET**, delivering features across multiple Agile sprint cycles.
- Debugged and analyzed legacy system architecture, identifying structural bottlenecks to improve maintainability and accelerate release cycles.
- Participated in cross-functional code reviews and daily standups to align software design with enterprise scalability requirements.

Research Experience

Adaptive Quantum-Chaotic Steganography (Ongoing) | *PyTorch, Qiskit, RL, Python* Jan 2025 – Present

- Designed an **AI-assisted BB84 steganography protocol** integrating Bernstein–Vazirani circuits and an **Improved Logistic Map (ILM)** for chaotic diffusion on BB84-derived keys.
- Trained a QBER classifier (1D CNN + attention / Transformer) achieving **AUROC 0.97** for detecting partial intercept-and-resend attacks on noisy IBM-style backends.
- Applied a **Deep Q-Learning** agent to tune ILM parameters (μ, γ), targeting higher NPCR/UACI with $\approx 1.5\%$ qubit overhead and ≈ 0.4 **ms** AI inference per epoch.

Technical Projects

FFmpeg WebGPU Integration | *Rust, WebGPU, WGSL, Vulkan, FFmpeg* Dec 2024 – Present

- Built a custom **WebGPU hwcontext** for FFmpeg enabling efficient GPU buffer management and zero-copy hardware-accelerated video pipelines.
- Ported existing **Vulkan**-based video processing filters to WebGPU, rewriting complex compute shaders into **WGSL** for broad cross-platform support.
- Leveraged **Rust** and the WGPU ecosystem to safely interface with FFmpeg's C-based architecture, optimizing memory ownership across CPU and GPU domains.

Ed25519 Firmware Signer | *Rust, ed25519-dalek, Argon2id, AES-256-GCM, libgit2* Jun 2024 – Present

- Designed a layered cryptographic pipeline: Ed25519 signing with private key encrypted at rest via **Argon2id** KDF + **AES-256-GCM**, ensuring keys are never written to disk in plaintext.
- Used `zeroize` to guarantee key material erasure from memory post-use and **OsRng** for CSPRNG-seeded key generation; enforced overwrite-protection prompts on key regeneration.
- Integrated **libgit2** for a git-tracked signing history, enabling auditable firmware release chains across embedded targets.

Leadership & Community

- **Linux Club:** Solved system-level and cryptographic challenges in a campus-wide **Capture The Flag (CTF)** competition.
- **OSPC Research Lead:** Mentored 50+ students on **Git**, collaborative development, and kernel-level basics at the Open Source Programming Club.